

Practice Questions: Functions in Python

1 Write a function to add two numbers

Input: 10, 20

Output: 30

2 Write a function to check if a number is even or odd

3 Write a function to find the square of a number

4 Write a function to print your name 5 times

5 Write a function to find the largest of two numbers

6 Write a function to find the factorial of a number

7 Write a function to check if a number is prime

8 Write a function to count vowels in a string

9 Write a function to reverse a string

10 Write a function to calculate the sum of elements in a list

1 1 Write a function to check if a string is a palindrome

1 2 Write a function to find the Fibonacci series up to n terms

1 3 Write a function that accepts a list and returns:

- Maximum
 - Minimum
 - Average
-

1 4 Write a function to count frequency of characters in a string

1 5 Write a function to check if a number is an Armstrong number

1 6 Function with no arguments and no return value

1 7 Function with arguments and no return value

1 8 Function with no arguments but return value

1 9 Function with arguments and return value

Q: Function to find factorial

◆ Create a menu-driven program using functions:

1. Add
 2. Subtract
 3. Multiply
 4. Divide
-

◆ Write a function that accepts multiple numbers using *args and returns their sum

1 Write a function to find the second largest element in a list

✗ Do NOT use sort()

2 Write a function to remove duplicate values from a list without using set()

3 Write a function to merge two sorted lists into a single sorted list

4 Write a function that rotates a list by k positions

Example:

Input → [1,2,3,4,5], k=2

Output → [3,4,5,1,2]

5 Write a function to find the missing number in a list from 1 to n

6 Write a recursive function to find GCD of two numbers

7 Write a recursive function to reverse a string

8 Write a recursive function to calculate power (x^n)

9 Write a function to check whether a number is a perfect number

10 Write a function to generate all prime numbers up to n

1 1 Write a function to find the longest palindrome in a string

1 2 Write a function to compress a string

Example:

Input → "aaabbc"

Output → "a3b2c1"

1 3 Write a function to check if a string follows title case rules

1 4 Write a function to find the first non-repeating character in a string

1 5 Write a function to remove punctuation from a string

1 6 Write a function using lambda to sort a list of tuples by second value

1 7 Write a function using map() to square all numbers in a list

1 8 Write a function using filter() to extract even numbers

1 9 Write a function using reduce() to find product of list elements

2 0 Write a function to validate a password

Rules:

- Minimum 8 characters
 - At least 1 uppercase
 - At least 1 digit
 - At least 1 special character
-

2 1 Write a function that simulates a bank account:

- deposit()
 - withdraw()
 - balance()
-

2 2 Write a function to parse a log file and count error messages

2 3 Write a function decorator that logs execution time of another function

2 4 Write a function to cache results (memoization) using a dictionary

2 5 Difference between:

- Function vs Method
 - return vs print
 - *args vs **kwargs
-

Design a function-based program for:

Student Management System

- Add student
- Update marks
- Calculate average
- Display topper